THE UMBRAL CALCULUS WITH DIVIDED DIFFERENCES

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Abstract. We trace the analogy between the classical umbral calculus of Rota, its lift to the ring of symmetric functions Λ via coalgebra morphisms and the Hall inner product (as developed in the author's masters thesis), and a prospective "divided difference umbral calculus" on the polynomial ring $\mathbb{Z}[x_1, \ldots, x_n]$. The column operator construction survives: Demazure operators play the role of generalized column operators, and key polynomials play the role of Littlewood–Richardson type bases. However, the parametric freedom collapses from an infinite-dimensional function space (delta series under composition) to the Macdonald parameters (q, t), due to the nilpotency and noncommutativity of divided differences. The Hecke algebra, the double affine Hecke algebra, and the Cherednik operators play the roles of the delta series, the umbral operators, and the column operators respectively.

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1. Introduction

The umbral calculus of Gian-Carlo Rota [10, 9] classifies the coalgebra automorphisms of the polynomial ring $\mathbb{Q}[x]$ by consideration of the algebra morphisms of the dual space $\mathbb{Q}[[y]]$. Every such automorphism is determined by an invertible formal power series $f(z)$ (a *delta series*), and gives rise to a *sequence of binomial type*: a basis $\{p_n(x)\}$ for $\mathbb{Q}[x]$ satisfying

$$p_n(x+x') = \sum_{k=0}^{n} \binom{n}{k} p_k(x)\, p_{n-k}(x').$$

In [5], the author lifted this machinery from $\mathbb{Q}[x]$ to the ring of symmetric functions Λ, using the coalgebra embedding $x^n/n! \mapsto h_n(X)$ and the Hall inner product. The result was a classification of all *Littlewood–Richardson type bases* $\{P_\lambda(X)\}$ of Λ: bases satisfying

$$P_\lambda(X+X') = \sum_{\mu,\nu} c^\lambda_{\mu\nu}\, P_\mu(X)\, P_\nu(X')$$

with the same structure constants $c^\lambda_{\mu\nu}$ as the Schur functions.

A natural question arises: what happens if we replace the differential operator $\mathcal{D} = d/dx$ not with the adjoint operators ∂_{h_n} on Λ, but with the *divided difference operators* ∂_i on the polynomial ring $\mathbb{Z}[x_1, \dots, x_n]$? This note traces the analogy, identifies what survives, what breaks, and what replaces the broken parts.

2. Three levels of the umbral calculus

2.1. **Level 1: The classical setting.** The polynomial ring $V = \mathbb{Q}[x]$ has a distinguished Hamel basis $\{1, x, x^2/2!, x^3/3!, \dots\}$ and a coalgebra structure given by:

$$\Delta\, p(x) = p(x+x').$$

The dual space $V^* = \mathbb{Q}[[y]]$ has basis $\{1, y, y^2, \dots\}$, and the pairing between V^* and V is:

$$(y^k, p(x)) = \left[\frac{x^k}{k!}\right] p(x) = \mathcal{L}\mathcal{D}^k\, p(x)$$

where $\mathcal{D} = d/dx$ is the differential operator and $\mathcal{L}$ is the constant term operator.

A *delta series* is a formal power series $f(y) \in \mathbb{Q}[[y]]$ with $f(0) = 0$ and $f'(0) \neq 0$, i.e., one which has a compositional inverse $g(y)$ satisfying $f(g(y)) = g(f(y)) = y$.

Proposition 2.1 ([5, Prop. 1.2])**.** *For every delta series $f(y)$ with compositional inverse $g(y)$, the* umbral operator

$$U_f = \sum_n \frac{x^n}{n!}\, \mathcal{L}\, f^n(\mathcal{D})$$

is the unique operator on $\mathbb{Q}[x]$ satisfying:

(1) $\mathcal{D} \circ U_f = U_f \circ g(\mathcal{D})$,
(2) $\mathcal{L} \circ U_f = \mathcal{L}$.

The group of delta series under composition — the *Jabotinsky group* (or Riordan group) — is infinite-dimensional. This is the source of the parametric freedom in the classical umbral calculus: there are infinitely many sequences of binomial type, one for each delta series.

2.2. **Level 2: Symmetric functions.** The coalgebra embedding

$$\frac{x^n}{n!} \mapsto h_n(X)$$

lifts the coproduct from $\mathbb{Q}[x]$ to Λ, where $\Delta f(X) = f(X + X')$ is given by plethystic addition of alphabets. The Hall inner product

$$\langle m_\lambda(Y), h_\mu(X)\rangle = \delta_{\lambda\mu}$$

replaces the pairing $(y^k, p(x)) = \mathcal{L}\mathcal{D}^k p(x)$. The role of the differential operator $\mathcal{D}$ is played by the adjoint operators ∂_{h_n}, defined by:

$$\langle \partial_f[g(Y)], h(X)\rangle = \langle g(Y),\, f(X)\, h(X)\rangle.$$

The *translation operator*

$$\mathcal{T}(z)[f(X)] = f(X + z) = \sum_n z^n\, \partial_{h_n}$$

and the *multiplication operator*

$$\mathcal{P}(z)[f(X)] = \Omega(Xz)\, f(X) = \sum_n z^n\, h_n(X) \cdot f(X)$$

combine to form the *column operator*:

$$C(z) = \mathcal{P}(-z)\, \mathcal{T}(1/z).$$

Proposition 2.2 ([5, Prop. 1.4])**.** *The Schur function has the column operator expression:*

$$s_\lambda(X) = C_{\lambda'_1} \circ C_{\lambda'_2} \circ \cdots \circ C_{\lambda'_m}[1].$$

The umbral operator on Λ is defined by $U_f[\Omega(Xz)] = \Omega_f(Xz) = \prod_x 1/(1 - f(z)x)$, extended multiplicatively. The generalized column operator is:

$$C_f(z) = U_f \circ C(z) \circ U_g$$

and the resulting bases

$$P_\lambda(X) = C^f_{\lambda'_1} \circ C^f_{\lambda'_2} \circ \cdots \circ C^f_{\lambda'_m}[1] = U_f[s_\lambda(X)] = \det(r_{\lambda_i+j-i}(X))$$

are of Littlewood–Richardson type, with dual bases $Q_\lambda(Y) = U^*_g[s_\lambda(Y)]$ satisfying:

$$\sum_\lambda P_\lambda(X)\, Q_\lambda(Y) = \Omega(XY) = \prod_{\substack{x\in X\\ y\in Y}} \frac{1}{1 - xy}.$$

The parametric freedom is the same as in the classical case: every delta series $f(y)$ gives a Littlewood–Richardson type basis.

2.3. **Level 3: Divided differences.** Now consider the polynomial ring $\mathbb{Z}[x_1, \ldots, x_n]$ with the *divided difference operators*:

$$\partial_i(f) = \frac{f - s_i(f)}{x_i - x_{i+1}}, \qquad i = 1, \ldots, n-1$$

where s_i is the simple transposition swapping x_i and x_{i+1}. These generate the *nil-Coxeter algebra* $\mathcal{N}_n$ with relations:

$$\partial_i^2 = 0, \tag{1}$$

$$\partial_i\partial_j = \partial_j\partial_i \quad \text{for } |i - j| \geq 2, \tag{2}$$

$$\partial_i\partial_{i+1}\partial_i = \partial_{i+1}\partial_i\partial_{i+1} \quad \text{(braid relation)}. \tag{3}$$

The *Demazure operators* (isobaric divided differences) are:

$$\pi_i(f) = \partial_i(x_i\, f) = \frac{x_i\, f - x_{i+1}\, s_i(f)}{x_i - x_{i+1}}.$$

These satisfy $\pi_i^2 = \pi_i$ (idempotence) and the same braid relations as the ∂_i.

3. The dictionary

The following table records the analogy across the three levels.

Classical $\mathbb{Q}[x]$	**Symmetric** Λ	**Divided difference** $\mathbb{Z}[x_1, \ldots, x_n]$
$\mathcal{D} = d/dx$	∂_{h_n}	$\partial_i = (f - s_i f)/(x_i - x_{i+1})$
$\mathcal{D}^n$	∂_{h_λ}	$\partial_w = \partial_{i_1} \cdots \partial_{i_k}$ (BGG)
$x^n/n!$	$h_n(X)$	$\mathfrak{S}_w(x)$ (Schubert polynomials)
$\mathcal{L}\mathcal{D}^k$ extracts $[x^k/k!]$	$\langle m_\lambda(Y), - \rangle$	∂_{w_0} as top extraction
$\Delta p = p(x + x')$	$\Delta f = f(X + X')$	no clean analogue (§5.1)
$\mathcal{T} = \sum x^n/n!\, \mathcal{L}\mathcal{D}^n$ (Taylor)	$\mathcal{T}(z) = \sum z^n \partial_{h_n}$	$f = \sum_w \partial_w(f)\|_0 \cdot \mathfrak{S}_w$
$\mathcal{P}(z) = e^{xz}\times$	$\mathcal{P}(z) = \Omega(Xz)\times$	—
$C(z) = \mathcal{P}(-z)\mathcal{T}(1/z)$	same	$\pi_i = x_i\partial_i$ (Demazure)
$p_n = U_f[x^n/n!]$	$P_\lambda = U_f[s_\lambda]$	$\kappa_\alpha = \pi_{i_1} \cdots \pi_{i_k}[x^\lambda]$
$\mathbb{Q}[\mathcal{D}]$ (commutative)	same	$\mathcal{N}_n$ (nil-Coxeter, dim $= n!$)
Delta series $f(y)$	same	Hecke parameter q
Jabotinsky group (∞-dim)	same	$\mathrm{Aut}(\mathcal{N}_n)$ (finite)
$\Omega(XY) = \sum s_\lambda s_\lambda$	same	$\prod_{i+j\le n+1}(x_i + y_j) = \sum \mathfrak{S}_w \mathfrak{S}_{w_0 w}$

4. What survives: the column operator construction

The column operator construction is the part of the umbral calculus that transfers most cleanly to divided differences.

4.1. **Column operators and Demazure operators.** In §2.2, the column operator $C(z) = \mathcal{P}(-z)\mathcal{T}(1/z)$ combines multiplication by a generating function with translation. The Demazure operator $\pi_i = x_i\partial_i$ has exactly this structure: it is multiplication by x_i followed by the divided difference ∂_i.

The Schur function is built by iterated column operators:

$$s_\lambda(X) = C_{\lambda'_1} \circ C_{\lambda'_2} \circ \cdots \circ C_{\lambda'_m}[1].$$

The key polynomial is built by iterated Demazure operators:

$$\kappa_\alpha = \pi_{i_1} \circ \pi_{i_2} \circ \cdots \circ \pi_{i_k}[x^\lambda]$$

where $s_{i_1} \cdots s_{i_k}$ is a reduced decomposition and λ is the partition obtained by sorting α.

4.2. **Generalized column operators.** In the symmetric function setting, the generalized column operator $C_f(z) = U_f \circ C(z) \circ U_g$ is conjugation of the Schur column operator by the umbral operators. The resulting bases $P_\lambda = C^f_{\lambda'_1} \circ \cdots \circ C^f_{\lambda'_m}[1]$ are of Littlewood–Richardson type.

The divided difference analogue of this conjugation replaces π_i with a *deformed Demazure operator*. The natural deformation is provided by the **Hecke algebra** $\mathcal{H}_n(t)$, which replaces $s_i^2 = 1$ with:

$$T_i^2 = (t-1)\, T_i + t. \tag{4}$$

The Hecke generator acts on the polynomial ring via the *Demazure–Lusztig operator*:

$$T_i = (t-1)\, \pi_i + s_i. \tag{5}$$

One may verify (4) directly using $\pi_i^2 = \pi_i$, $s_i^2 = \mathrm{id}$, $s_i\pi_i = \pi_i$, and $\pi_i s_i = \mathrm{id} + s_i - \pi_i$:

$$\begin{aligned} T_i^2 &= (t-1)^2\pi_i + (t-1)(\pi_i s_i + s_i\pi_i) + \mathrm{id} \\ &= (t-1)^2\pi_i + (t-1)(\mathrm{id} + s_i) + \mathrm{id} \\ &= (t-1)^2\pi_i + (t-1)s_i + t \\ &= (t-1)T_i + t. \qquad \checkmark \end{aligned}$$

The formula (5) makes the role of the parameter t transparent: it controls the mixture of the Demazure projection π_i and the reflection s_i.

Parameter	**Operator**	**Polynomials**
$t = 1$	$T_i = s_i$ (reflection)	Schur / symmetric functions
$t = 0$	$T_i = s_i - \pi_i$ (0-Hecke)	key polynomials / atoms
$t = -1$	$T_i = -2\pi_i + s_i$	—
general t	$T_i = (t-1)\pi_i + s_i$	Hall–Littlewood / Macdonald

Remark 4.1. At $t = 0$, the relation becomes $T_i^2 = -T_i$, so $T_i + 1$ is idempotent: $(T_i + 1)^2 = T_i + 1$. This is the **0-Hecke algebra**, which is the algebra of Demazure operators. Note that the *nil-Coxeter algebra* ($\partial_i^2 = 0$) is a *different* quotient of the braid group algebra — it is **not** a specialization of the Hecke algebra at any value of t. The divided difference ∂_i requires both s_i and the polynomial ring for its definition: $\partial_i = (1 - s_i)/(x_i - x_{i+1})$. It is an element of the *degenerate affine Hecke algebra*, not the finite Hecke algebra.

4.3. **Anti-Demazure operators and atoms.** The anti-Demazure operator

$$\bar{\pi}_i = \pi_i - \mathrm{id}$$

satisfies $\pi_i + \bar{\pi}_i = \mathrm{id}$, so the two operators are complementary projections. Since $\pi_i = \mathrm{id} + \bar{\pi}_i$, expanding a product of Demazure operators yields:

$$\kappa_\alpha = \sum_{\beta \le \alpha} A_\beta$$

where the *Demazure atoms* $A_\alpha = \bar{\pi}_{i_1} \circ \cdots \circ \bar{\pi}_{i_k}[x^\lambda]$ are obtained by using $\bar{\pi}$ at every step. This decomposition is the direct analogue of the expansion $P_\lambda(X) = \sum_{\nu \subseteq \lambda} \square_{\lambda\nu}\, s_\nu(X)$ in §2.2, where the Schur expansion coefficients arise from the "identity vs. umbral" choice at each step.

5. What breaks: three structural obstructions

5.1. **The coproduct does not lift.** The coproduct $\Delta f(X) = f(X+X')$ on Λ, given by plethystic addition of alphabets, is the engine of the classical theory. It gives rise to the binomial-type identity $p_n(x+x') = \sum \binom{n}{k} p_k\, p_{n-k}$ and its symmetric function generalization $P_\lambda(X+X') = \sum c^\lambda_{\mu\nu} P_\mu P_\nu$.

The polynomial ring $\mathbb{Z}[x_1, \dots, x_n]$ has variable-wise addition $\Delta p(x) = p(x_1 + y_1, \dots, x_n + y_n)$, but Schubert polynomials do *not* satisfy a binomial-type property under this coproduct. That is, there is no identity of the form:

$$\mathfrak{S}_w(x+y) = \sum_{u,v} c^w_{uv}\, \mathfrak{S}_u(x)\, \mathfrak{S}_v(y).$$

The "Cauchy identity" for Schubert polynomials is instead a *finite product*:

$$\prod_{i+j \le n+1} (x_i + y_j) = \sum_{w \in \mathfrak{S}_n} \mathfrak{S}_w(x)\, \mathfrak{S}_{w_0 w}(y)$$

which represents the class of a point in the cohomology ring $H^*(\mathrm{Fl}(n) \times \mathrm{Fl}(n))$. This is Poincaré duality, not a Cauchy kernel — it is a polynomial, not a formal power series.

Remark 5.1. The theory of *back-stable Schubert polynomials* (Lam–Lee–Shimozono [6]) embeds Schubert polynomials into a larger ring that *does* have a good coproduct, partially recovering the coalgebra structure. This may be the right setting for a full divided-difference umbral calculus.

5.2. **The operator algebra is noncommutative.** The umbral operator $U_f = \sum x^n/n!\, \mathcal{L} f^n(\mathcal{D})$ relies on $\mathcal{D}$ generating a commutative algebra $\mathbb{Q}[\mathcal{D}]$. The delta series $f(y)$ and its compositional inverse $g(y)$ live in the group of invertible formal power series under composition — the Jabotinsky (or Riordan) group, which is infinite-dimensional.

The divided differences $\partial_1, \dots, \partial_{n-1}$ generate the nil-Coxeter algebra $\mathcal{N}_n$, which is:

- **noncommutative** (the braid relation is not commutativity),
- **finite-dimensional** ($\dim \mathcal{N}_n = n!$, with basis $\{\partial_w : w \in \mathfrak{S}_n\}$),
- **nilpotent** ($\partial_i^2 = 0$).

The finiteness and noncommutativity mean that "formal power series in ∂" do not form a commutative ring, and compositional inversion of delta series has no direct analogue. The entire Jabotinsky group collapses to the automorphism group of $\mathcal{N}_n$, which is finite.

5.3. **Lowering happens in multiple directions.** In $\mathbb{Q}[x]$, the operator $\mathcal{D}$ lowers degree by 1 along a single axis. Taylor's formula $\mathcal{I} = \sum x^n/n!\, \mathcal{L}\mathcal{D}^n$ expresses the identity as a sum over $\mathbb{N}$.

With divided differences, there are $n-1$ operators $\partial_1, \dots, \partial_{n-1}$, each lowering degree by 1 but in a different root direction. The Taylor formula becomes the *Schubert expansion*:

$$f = \sum_{w \in \mathfrak{S}_n} \partial_w(f)\big|_{x=0} \cdot \mathfrak{S}_w(x)$$

which is a sum over the symmetric group $\mathfrak{S}_n$ rather than over $\mathbb{N}$. The multi-directionality is intrinsic to the theory and cannot be reduced to a single operator.

6. What replaces the broken parts

6.1. **The Hecke parameter replaces delta series.** In the classical theory, the parametric family of umbral operators is indexed by delta series $f(y)$ — an infinite-dimensional space. In the divided difference setting, the natural deformation is the Hecke algebra $\mathcal{H}_n(t)$, which replaces the symmetric group relation $s_i^2 = 1$ with (4).

As shown in Remark 4.1, $t = 0$ gives the 0-Hecke algebra (the algebra of Demazure operators, with key polynomials and atoms), while $t = 1$ recovers the symmetric group (with Schur polynomials). At generic t, the Demazure–Lusztig operators generate Hall–Littlewood polynomials.

To obtain the full Macdonald polynomials, one needs the second parameter q, which arises from the *double affine Hecke algebra* (see §6.2 below).

Remark 6.1. The collapse from an infinite-dimensional parameter space (delta series) to a two-dimensional one (Macdonald's (q,t)) reflects the collapse from $\mathbb{Q}[\mathcal{D}] \cong \mathbb{Q}[[y]]$ (infinite-dimensional) to the finite-dimensional operator algebras generated by divided differences. The braid group algebra admits only finitely many interesting quotients, so only finitely many deformation parameters survive.

6.2. **The DAHA and Cherednik operators.** The single Hecke parameter t gives Hall–Littlewood polynomials. To obtain the full Macdonald polynomials $P_\lambda(x;q,t)$, one needs the *double affine Hecke algebra* (DAHA), introduced by Cherednik [1]. The DAHA $\ddot{\mathcal{H}}_n(q,t)$ is generated by three families of operators:

- $T_1, \dots, T_{n-1}$ (Hecke generators, parameter t),
- $x_1, \dots, x_n$ (polynomial multiplication),
- $Y_1, \dots, Y_n$ (Cherednik operators),

subject to the cross-relations:

$$T_i\, x_i\, T_i = t\, x_{i+1}, \tag{6}$$

$$T_i\, Y_i\, T_i = Y_{i+1}. \tag{7}$$

The Cherednik operators commute:

$$Y_i\, Y_j = Y_j\, Y_i \qquad \text{for all } i,j. \tag{8}$$

The *nonsymmetric Macdonald polynomials* $E_\alpha(x;q,t)$ are the simultaneous eigenfunctions of the Cherednik operators:

$$Y_i\, E_\alpha = \xi_i(\alpha;q,t) \cdot E_\alpha$$

where $\xi_i(\alpha;q,t)$ are explicit eigenvalues depending on the composition α and the parameters.

6.2.1. *The Cherednik operators as umbral operators.* In [5], the umbral operator U_f is characterized by the intertwining property:

$$\mathcal{D} \circ U_f = U_f \circ g(\mathcal{D}).$$

This says that U_f *conjugates* the derivative into a new operator $g(\mathcal{D})$. The Cherednik operators satisfy an analogous intertwining relation (7): the Hecke generator T_i permutes the Cherednik operators, just as U_f transforms the derivative.

More precisely, the analogy is:

Umbral calculus on Λ	**DAHA**
Umbral operator U_f	Cherednik operator Y_i
$\mathcal{D} \circ U_f = U_f \circ g(\mathcal{D})$	$T_i\, Y_i\, T_i = Y_{i+1}$
Delta series $f(y)$	Parameters (q,t)
Jabotinsky group (∞-dim)	Two parameters
$U_f[s_\lambda] = P_\lambda$	$Y_i\, E_\alpha = \xi_i \cdot E_\alpha$
$\Omega(XY) = \sum P_\lambda Q_\lambda$	Macdonald Cauchy identity

6.2.2. *Intertwining operators as column operators.* The closest analogue of the column operators $C_f(z) = U_f \circ C(z) \circ U_g$ from [5] is the family of *Knop–Sahi intertwining operators* ϕ_i in the DAHA [4, 11]. These satisfy:

$$Y_i \, \phi_i = \phi_i \, Y_{i+1} \tag{9}$$

and build up the Macdonald polynomials step by step:

$$E_{s_i \cdot \alpha} \propto \phi_i(E_\alpha)$$

starting from a "base" polynomial. This is directly analogous to the column operator construction

$$s_\lambda(X) = C_{\lambda'_1} \circ C_{\lambda'_2} \circ \cdots \circ C_{\lambda'_m}[1]$$

from [5, Prop. 1.4]: the intertwining operators are the "generalized column operators" of the Hecke world.

6.2.3. *The Macdonald involution as compositional inverse.* In [5], the umbral operator U_f and its adjoint U_g^* are related by the compositional inverse $g = f^{-1}$: the LR-type basis $P_\lambda = U_f[s_\lambda]$ has dual $Q_\lambda = U_g^*[s_\lambda]$, and the duality is controlled by $f \leftrightarrow g$.

In the Macdonald setting, the analogous duality is the *Macdonald involution*: the symmetry that sends $P_\lambda(x; q, t)$ to $Q_\lambda(x; q, t)$. This involution involves interchanging the roles of the two parameters (roughly $q \leftrightarrow t^{-1}$, up to normalizations). The compositional inverse $g = f^{-1}$ of the delta series — which drives the entire duality theory in [5] — becomes the (q, t)-duality of Macdonald polynomials.

6.3. **The nilHecke algebra replaces the operator algebra.** The *nilHecke algebra* $\mathcal{NH}_n$ is the smash product $\mathbb{Z}[x_1, \ldots, x_n] \rtimes \mathcal{N}_n$, with the cross-relation:

$$\partial_i \, x_i = 1 + x_{i+1} \, \partial_i. \tag{10}$$

This is the divided-difference Leibniz rule, and it plays the role of the commutation relation $\mathcal{D} \circ U_f = U_f \circ g(\mathcal{D})$ from [5, Prop. 1.2]. The nilHecke algebra is the single algebra in which both "multiplication" ($\mathcal{P}$) and "translation" ($\mathcal{T}$) operators naturally live.

Remark 6.2. In the symmetric function setting, the translation and multiplication operators satisfy the commutation relation (Lemma 1.1 of [5]):

$$\mathcal{T}(v) \, \mathcal{P}(w) = \Omega(vw) \, \mathcal{P}(w) \, \mathcal{T}(v).$$

The nilHecke relation (10) is the pointwise, operator-level version of this: ∂_i (a single step of translation) and x_i (multiplication by a single variable) do not commute, and their commutator is controlled by s_i.

6.4. **The Yang–Baxter equation replaces the binomial theorem.** The binomial-type identity $p_n(x + x') = \sum \binom{n}{k} p_k(x) \, p_{n-k}(x')$ is the defining property of sequences of binomial type. In the divided difference setting, the structural identity that replaces it is the *Yang–Baxter equation*.

Following Fomin and Kirillov [2], define the *Yang–Baxter exponential* in the nil-Coxeter algebra:

$$\mathfrak{E}(z) = \prod_i (1 + z \, u_i)$$

where u_i are the nil-Coxeter generators. This satisfies:

$$(1 + z \, u_i)(1 + z \, u_{i+1})(1 + z \, u_i) = (1 + z \, u_{i+1})(1 + z \, u_i)(1 + z \, u_{i+1}).$$

The Yang–Baxter equation is the "binomial theorem" for divided differences: it governs how the exponential factors when the "alphabet" is split. Just as the binomial theorem for p_n is equivalent to $\Delta p_n = \sum \binom{n}{k} p_k \otimes p_{n-k}$ being a coalgebra map, the Yang–Baxter equation is equivalent to a *quasi-triangularity* condition on the nil-Coxeter algebra.

6.5. **Poincaré duality replaces the Cauchy kernel.** The Cauchy kernel $\Omega(XY) = \prod 1/(1-xy) = \sum_\lambda P_\lambda(X)\,Q_\lambda(Y)$ characterizes dual bases in Λ with respect to the Hall inner product. In the divided difference setting, the "Cauchy identity" is the finite product:

$$\prod_{i+j\le n+1}(x_i+y_j) = \sum_{w\in\mathfrak{S}_n} \mathfrak{S}_w(x)\,\mathfrak{S}_{w_0w}(y).$$

The dual of $\mathfrak{S}_w$ is $\mathfrak{S}_{w_0w}$, where $\ell(w)+\ell(w_0w)=\ell(w_0)=\binom{n}{2}$, reflecting Poincaré duality on the flag variety $\mathrm{Fl}(n)$.

Symmetric functions	**Schubert calculus**
$\mathrm{Rep}(GL_n)$	$H^*(\mathrm{Fl}(n))$
Cauchy kernel (power series)	finite product (polynomial)
s_λ self-dual	$\mathfrak{S}_{w_0w}$ dual to $\mathfrak{S}_w$
$\sum_\lambda s_\lambda s_\lambda = \Omega$	$\sum_w \mathfrak{S}_w\mathfrak{S}_{w_0w} = \prod(x_i+y_j)$
LR coefficients	Schubert structure constants

7. Key polynomials and Demazure atoms as an LR-type pair

The Cauchy identity for key polynomials and Demazure atoms is:

$$\prod_{i,j=1}^{n}\frac{1}{1-x_iy_j} = \sum_{\alpha\in\mathrm{Comp}_n} \kappa_\alpha(x_1,\dots,x_n)\cdot A_{w_0\cdot\alpha}(y_1,\dots,y_n)$$

where $w_0\cdot\alpha$ reverses the composition. This is the closest analogue of the symmetric function identity $\sum_\lambda P_\lambda(X)\,Q_\lambda(Y) = \Omega(XY)$ from [5]: the key polynomials play the role of P_λ and the Demazure atoms play the role of Q_λ.

The decomposition $\kappa_\alpha = \sum_{\beta\le\alpha} A_\beta$ is upper-triangular, making key polynomials and atoms dual bases — exactly as the Littlewood–Richardson type bases P_λ and their duals Q_λ are related by an upper-triangular change of basis from the Schur functions.

8. The collapse of parametric freedom

Question 8.1. In the classical umbral calculus, the Jabotinsky group of delta series parametrizes all sequences of binomial type. What is the "Jabotinsky group" for divided differences?

The answer appears to be: *essentially trivial*. The automorphism group of the nil-Coxeter algebra $\mathcal{N}_n$ is generated by the diagram automorphisms of the Dynkin diagram A_{n-1} (i.e., the map $\partial_i \mapsto \partial_{n-i}$, corresponding to $w \mapsto w_0ww_0$). This is a finite group, in stark contrast to the infinite-dimensional Jabotinsky group.

The Hecke deformation $s_i^2 = 1 \;\rightsquigarrow\; T_i^2 = (t-1)T_i + t$ provides a one-parameter family, and the DAHA (§6.2) extends this to the two-parameter Macdonald (q,t)-family. But these are *finitely many* parameters, not the infinite-dimensional family of delta series.

Conjecture 8.2. *The Macdonald (q,t)-parameters exhaust the deformation freedom available when the derivative is replaced by divided differences in the umbral calculus. More precisely, any "divided-difference umbral operator" that produces a basis of $\mathbb{Z}[x_1,\dots,x_n]$ with Littlewood–Richardson type structure constants is equivalent, up to automorphisms of $\mathcal{N}_n$, to a specialization of the Macdonald construction.*

9. Directions

(1) **Back-stable Schubert polynomials.** The work of Lam–Lee–Shimozono [6] embeds Schubert polynomials into a ring with a good coproduct. Does this restore enough coalgebra structure to support a full umbral calculus?
(2) **Noncommutative symmetric functions.** The theory of Gelfand–Krob–Lascoux–Leclerc–Thibon–Retakh [3] provides a Hopf algebra of noncommutative symmetric functions with divided differences built in. Is this the "right" home for the divided-difference umbral calculus?
(3) **The nilHecke algebra as a bialgebra.** The nilHecke algebra $\mathcal{NH}_n$ has both a product (algebra structure) and a coproduct (induced from the inclusion $\mathfrak{S}_m \times \mathfrak{S}_k \hookrightarrow \mathfrak{S}_{m+k}$). Does the resulting bialgebra structure support a notion of "binomial type"?
(4) **Elliptic deformations.** Beyond the Hecke parameter q and the Macdonald parameters (q, t), elliptic deformations (Ruijsenaars, Rains) provide further parameters. Are these additional "delta series" in the divided-difference setting?
(5) **Categorification.** The nilHecke algebra categorifies the quantum group $U_q(\mathfrak{sl}_n)$ via Khovanov–Lauda–Rouquier algebras. Does the umbral calculus have a categorical shadow?

References

[1] I. Cherednik, *Double affine Hecke algebras*, London Math. Soc. Lecture Note Series **319**, Cambridge University Press, 2005.
[2] S. Fomin and A. N. Kirillov, *The Yang–Baxter equation, symmetric functions, and Schubert polynomials*, Discrete Math. **153** (1996), 123–143.
[3] I. M. Gelfand, D. Krob, A. Lascoux, B. Leclerc, V. Retakh, and J.-Y. Thibon, *Noncommutative symmetric functions*, Adv. Math. **112** (1995), 218–348.
[4] F. Knop, *Integrality of two variable Kostka functions*, J. Reine Angew. Math. **482** (1997), 177–189.
[5] R. Langer, *Symmetric functions and Macdonald polynomials*, Masters thesis, University of Melbourne, 2008.
[6] T. Lam, S. J. Lee, and M. Shimozono, *Back stable Schubert calculus*, Compositio Math. **157** (2021), 883–962.
[7] I. G. Macdonald, *Symmetric functions and Hall polynomials*, 2nd ed., Oxford University Press, 1995.
[8] V. Reiner and M. Shimozono, *Key polynomials and a flagged Littlewood–Richardson rule*, J. Combin. Theory Ser. A **70** (1995), 107–143.
[9] S. Roman, *The Umbral Calculus*, Academic Press, 1984.
[10] G.-C. Rota, *Finite operator calculus*, Academic Press, 1975.
[11] S. Sahi, *Interpolation, integrality, and a generalization of Macdonald's polynomials*, Internat. Math. Res. Notices **1996**, no. 10, 457–471.